4 Power Question 2018: Partitions

Instructions: The power question is worth 50 points; each part's point value is given in brackets next to the part. To receive full credit, the presentation must be legible, orderly, clear, and concise. If a problem says "list" or "compute," you need not justify your answer. If a problem says "determine," "find," or "show," then you must show your work or explain your reasoning to receive full credit, although such explanations do not have to be lengthy. If a problem says "justify" or "prove," then you must prove your answer rigorously. Even if not proved, earlier numbered items may be used in solutions to later numbered items, but not vice versa. Pages submitted for credit should be NUMBERED IN CONSECUTIVE ORDER AT THE TOP OF EACH PAGE in what your team considers to be proper sequential order. PLEASE WRITE ON ONLY ONE SIDE OF THE ANSWER PAPERS. Put the TEAM NUMBER (not the team name) on the cover sheet used as the first page of the papers submitted. Do not identify the team in any other way.

Binary Partitions

A *binary partition* of a positive integer n is an ordered n-tuple of non-increasing integers, each of which is either 0 or a power of 2, whose sum is n. Each of the integers in the n-tuple is called a *part* of the partition. Each binary partition of n has n parts. Let $p_2(n)$ denote the number of binary partitions of n. For example, $p_2(3) = 2$ because of the two ordered triples $(2, 1, 0)$ and $(1, 1, 1)$.

1. Compute $p_2(n)$ for $n = 4$, 5, 6, and 7. [4 pts]

2. **a.** Show that $p_2(n) \leq p_2(n+1)$ for all positive integers n. [3 pts]

b. Is the inequality strict for sufficiently large n? Justify your answer. [3 pts]

3. **a.** Prove that if n is even and $n \geq 4$, then $p_2(n) = p_2\left(\frac{n}{2}\right) + p_2(n-2)$. [2 pts]

b. Find the least $n > 1$ such that $p_2(n)$ is odd, or prove that no such n exists. [2 pts]

Partial Orderings

A *partial ordering* on a set S is a relation, usually denoted $\preceq$, such that all of the following conditions are true:

- $a \preceq a$ for all $a \in S$ **(reflexivity property)**,
- $a \preceq b$ and $b \preceq c$ implies $a \preceq c$ for all $a, b, c \in S$ **(transitivity property)**, and
- $a \preceq b$ and $b \preceq a$ implies $a = b$ for all $a, b \in S$ **(antisymmetry property)**.

The word "partial" refers to the possibility that some two elements, a and b, may be *incomparable*, i.e., neither $a \preceq b$ nor $b \preceq a$ (these negative relations on $\preceq$ are sometimes written as $a \npreceq b$ and $b \npreceq a$, respectively). This Power Question largely defines and explores a partial ordering on the set of binary partitions of n.

The notation $a \prec b$ means that $a \preceq b$ and $a \neq b$. The symbols $\succ$ and $\succeq$ may also be used, and they are defined in the following way: $a \succ b$ means $b \prec a$, and $a \succeq b$ means $b \preceq a$.

If a and b are elements of S with $a \prec b$, and if there is no element $c \in S$ for which $a \prec c \prec b$, then it is said that b *covers* a.

Let a denote the binary partition $(a_1, a_2, \ldots, a_n)$. Similarly, let b denote the binary partition $(b_1, b_2, \ldots, b_n)$. In this Power Question, define $a \prec b$ if it is possible to obtain a from b by a sequence of replacing one 2^k by two 2^{k-1}s (and deleting a 0). For example, $(4, 1, 1, 1, 1, 0, 0, 0) \prec (4, 4, 0, 0, 0, 0, 0, 0)$ because $(4, 1, 1, 1, 1, 0, 0, 0) \prec (4, 2, 1, 1, 0, 0, 0, 0) \prec (4, 2, 2, 0, 0, 0, 0, 0) \prec (4, 4, 0, 0, 0, 0, 0, 0)$. This Power Question will use this partial ordering on binary partitions.

4. **a.** Show that the binary partitions of 5 are totally ordered; i.e., if p and p' are two different binary partitions of 5, then either $p \prec p'$ or $p' \prec p$. [2 pts]

b. Show that the binary partitions of 8 are not totally ordered, i.e., find two binary partitions of 8 – call them q and q' – such that $q \not\prec q'$ and $q' \not\prec q$. [3 pts]

5. **a.** Find the smallest binary partition of n, using this partial ordering. That is, find the binary partition p such that for all other binary partitions p', $p \prec p'$. [2 pts]

b. Find the largest binary partition of n, using this partial ordering. That is, find the binary partition P such that for all other binary partitions P', $P \succ P'$. [3 pts]

Hasse Diagrams

Suppose that a set S has a partial ordering $\preceq$. Then a *Hasse diagram* can be used to display the covering relation in a graphical way. The Hasse diagram is a graph whose vertices are the elements of S and where edges are drawn between two elements x and y if $x \prec y$ or $y \prec x$ and if there is no element z for which $x \prec z$ and $z \prec y$ or for which $y \prec z$ and $z \prec x$. Also, y appears "above" x if $x \prec y$. Note that it is possible for more than one element to appear on the same level of a Hasse diagram. For example, the partially ordered set of divisors of 12, ordered by divisibility, is shown in Figure 1. In this diagram, the number 1 is said to be at Level 0 because it is the least divisor in the partial ordering shown. The numbers 2 and 3 are said to be on Level 1 because $2 \succ 1$ and $3 \succ 1$ and there is no number n such that $2 \succ n$ and $n \succ 1$ or $3 \succ n$ and $n \succ 1$. Other Levels are similarly defined. The number 12 is on Level 3.

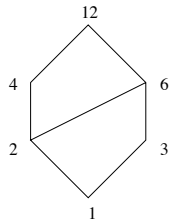


Figure 1

Hasse diagrams can be drawn to show the ordering of partitions such as the ones from Problems 4 and 5. For convenience, rather than labeling the vertices in the Hasse diagram with the partition itself, like $(8, 0, 0, 0, 0, 0, 0, 0)$, it is common to label the vertex with its nonzero parts only, using exponents to indicate parts within the partition with multiplicity greater than 1. For example, the partition $(8, 0, 0, 0, 0, 0, 0, 0)$ would be labeled 8, the partition $(4, 2, 2, 0, 0, 0, 0, 0)$ would be labeled 42^2, and the partition $(2, 2, 1, 1, 1, 1, 0, 0)$ would be labeled $2^2 1^4$.

6. **a.** Draw the Hasse diagram for the binary partitions of 8. Label each vertex. [1 pt]

b. List a path through the Hasse diagram for the binary partitions of 8 that begins at the bottom vertex (that is, the vertex at Level 0) and, traveling only along edges, passes through every other vertex exactly once. Such a path is called a *Hamiltonian path*. [1 pt]

c. Let n be an even integer with $n > 4$. Let S be the set of binary partitions of n. Let S_1 be the set of binary partitions of $\frac{n}{2}$. Let S_2 be the set of binary partitions of $n - 2$. Prove that there is a bijection B (i.e., a one-to-one correspondence) from the set S to the set $S_1 \cup S_2$. Prove that this bijection B preserves order; that is, given that $p \prec p'$ for binary partitions $p, p' \in S$, then either $B(p) \prec B(p')$ or $B(p)$ and $B(p')$ are incomparable. [3 pts]

d. Prove that for each positive integer n, the Hasse diagram of the binary partitions of n has a Hamiltonian path that begins with the vertex at Level 0. [5 pts]

Let $f_L(n)$ represent the number of elements at level L in the Hasse diagram of the binary partitions of n.

7. Prove that the value of $f_L(n)$ is the number of binary partitions of n that have $n - L$ nonzero parts. [3 pts]

8. Not all partitions are binary partitions. Some partitions have parts of the form $2^j - 1$ where j is a nonnegative integer. Such partitions will be called *s-partitions*, and their parts are written in nonincreasing order. Two s-partitions of 5 are $(3, 1, 1, 0, 0)$ and $(1, 1, 1, 1, 1)$.

a. List two partitions of 7, one that is an s-partition and one that is neither an s-partition nor a binary partition. Make sure to identify which is which. [2 pts]

b. Prove that if $n \geq 2L$, the value of $f_L(n)$ is equal to the number of s-partitions of L. [3 pts]

Trinary Partitions

A *trinary partition* of a positive integer n is an ordered n-tuple of non-increasing integers, each of which is either 0 or a power of 3, whose sum is n. Let $p_3(n)$ denote the number of trinary partitions of n. For example, $p_3(4) = 2$ because of the two ordered quadruples $(3, 1, 0, 0)$ and $(1, 1, 1, 1)$.

As with binary partitions, one can define partial orderings for trinary partitions. Let c denote the trinary partition $(c_1, c_2, \ldots, c_n)$. Similarly, let d denote the trinary partition $(d_1, d_2, \ldots, d_n)$. Define $c \prec d$ if it is possible to obtain c from d by a sequence of replacing one 3^k by three 3^{k-1}s (and deleting two 0s).

9. Draw the Hasse diagram for the trinary partitions of 12. Label each vertex. [2 pts]

10. State a value of n less than 23 for which the Hasse diagram of the trinary partitions of n does **not** contain a Hamiltonian path. Prove your claim. (Recall that a Hamiltonian path is defined in Problem 6b.) [6 pts]